Mouse-level RNA TE differential analysis

20 mice | 22,251 provisional Microglia-like RNA cells | fractional expected allocations

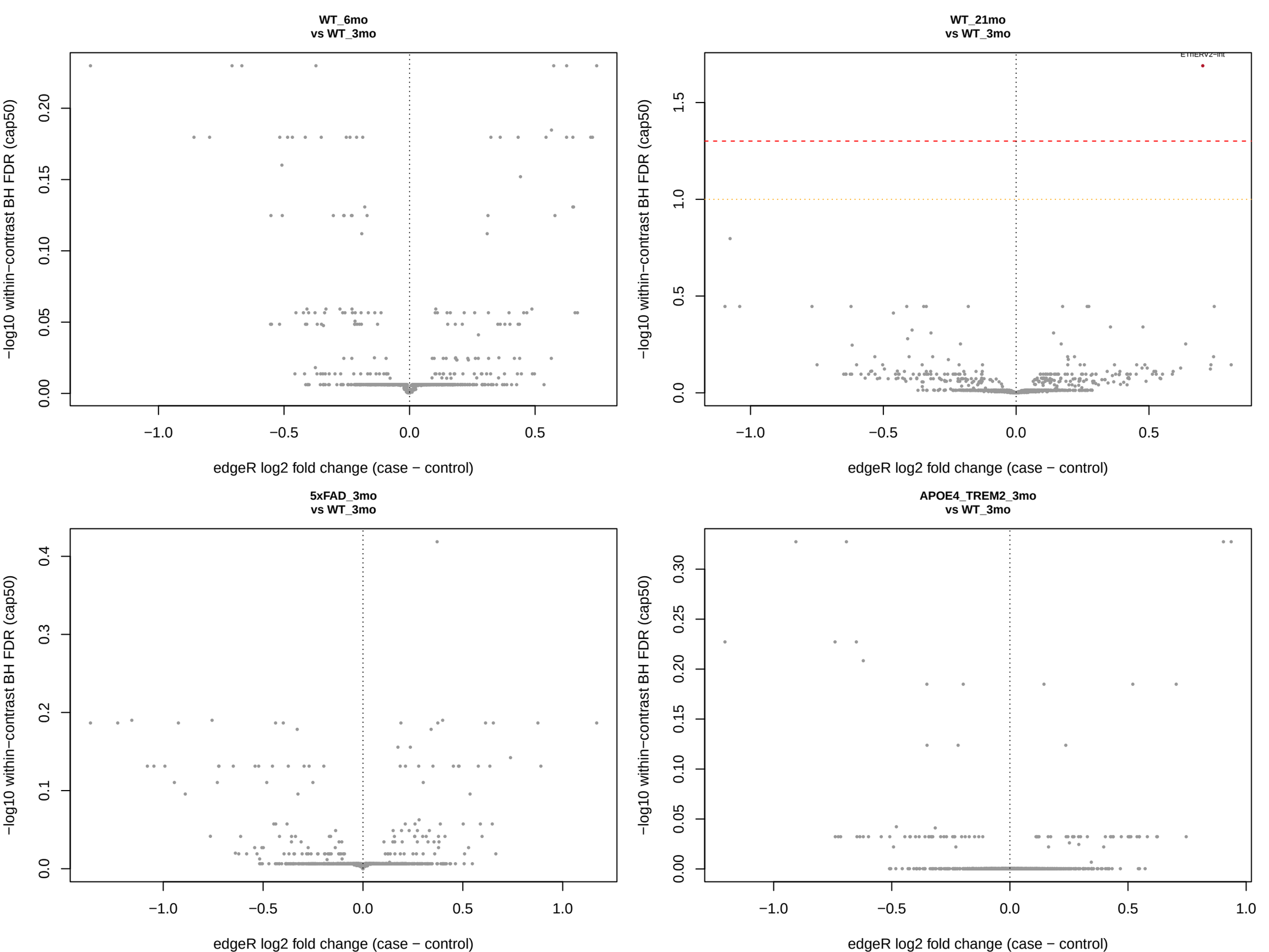
Primary: edgeR v4 robust QL, condition + sex, TMM-relative TE composition. No absolute TE-induction claim.

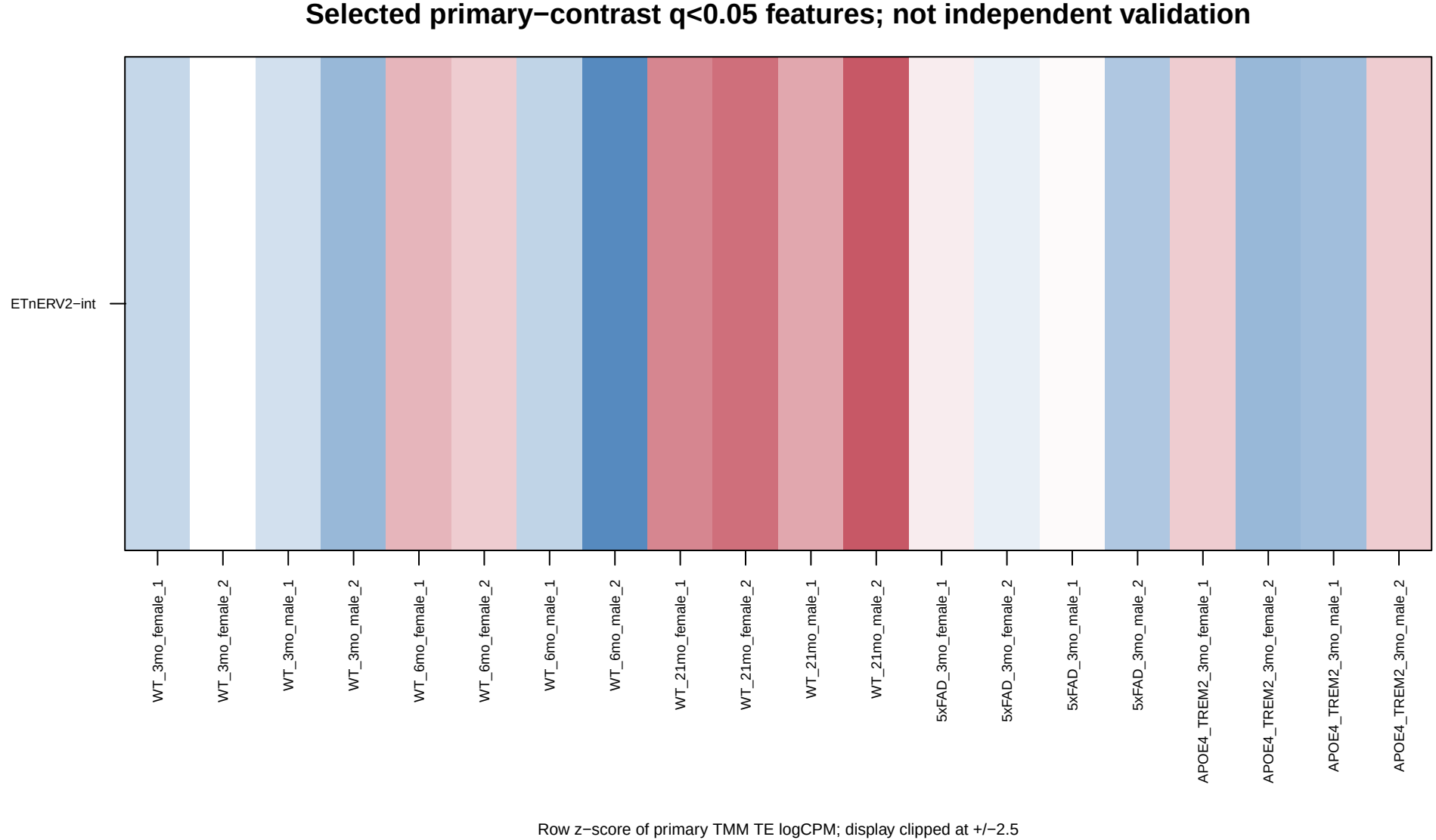
Level	Case vs control	Scope	Eligible	q<.05	q<.10	Pooled4 q<.05	Pooled10 q<.05
Subfamily	WT_6mo vs WT_3mo	Primary	785	0	0	0	0
Subfamily	WT_21mo vs WT_3mo	Primary	785	1	1	0	0
Subfamily	5xFAD_3mo vs WT_3mo	Primary	785	0	0	0	0
Subfamily	APOE4_TREM2_3mo vs WT_3mo	Primary	785	0	0	0	0
Subfamily	WT_21mo vs WT_6mo	Secondary	785	0	0	NA	0
Subfamily	5xFAD_3mo vs WT_6mo	Secondary	785	0	0	NA	0
Subfamily	APOE4_TREM2_3mo vs WT_6mo	Secondary	785	0	0	NA	0
Subfamily	5xFAD_3mo vs WT_21mo	Secondary	785	0	0	NA	0
Subfamily	APOE4_TREM2_3mo vs WT_21mo	Secondary	785	0	1	NA	0
Subfamily	APOE4_TREM2_3mo vs 5xFAD_3mo	Secondary	785	0	0	NA	0
Raw family	WT_6mo vs WT_3mo	Primary	38	0	0	0	0
Raw family	WT_21mo vs WT_3mo	Primary	38	0	0	0	0
Raw family	5xFAD_3mo vs WT_3mo	Primary	38	0	0	0	0
Raw family	APOE4_TREM2_3mo vs WT_3mo	Primary	38	0	0	0	0
Raw family	WT_21mo vs WT_6mo	Secondary	38	0	0	NA	0
Raw family	5xFAD_3mo vs WT_6mo	Secondary	38	0	0	NA	0
Raw family	APOE4_TREM2_3mo vs WT_6mo	Secondary	38	0	0	NA	0
Raw family	5xFAD_3mo vs WT_21mo	Secondary	38	0	0	NA	0
Raw family	APOE4_TREM2_3mo vs WT_21mo	Secondary	38	0	0	NA	0
Raw family	APOE4_TREM2_3mo vs 5xFAD_3mo	Secondary	38	0	0	NA	0

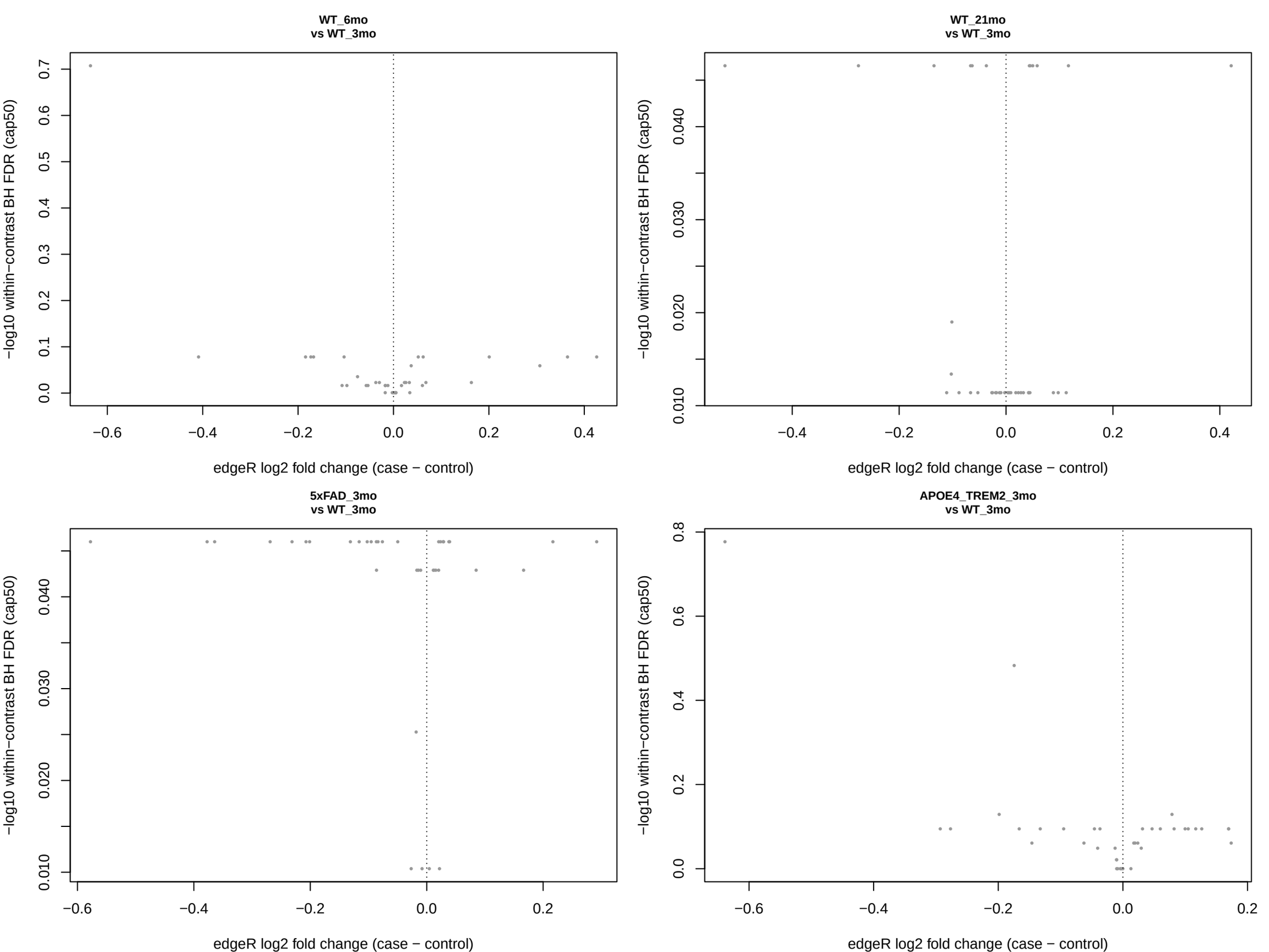
q columns use BH within each contrast; q<0.10 is exploratory. Pooled4 and pooled10 are separate multiplicity sensitivities within each feature level. Secondary contrasts can mix age and genotype; consult the confounding flags. Counts above are the actual reported calls, not evidence that a preferred normalization was selected by discoveries. Separate voom marginal moderated 95% intervals are sensitivity results, not QL or simultaneous intervals. EM assignment uncertainty is not propagated. Raw family conflicts stay unsplit. No cell-level replication, joint RNA-ATAC embedding or autonomous-TE claim.

Following pages: four primary-contrast volcanoes and selected q<0.05 heatmaps for both feature levels.

Zero-hit heatmap diagnostics are retained. All ten contrasts, diagnostic PDFs and full result TSVs are in the accompanying ZIP.


WT_6mo
vs WT_3mo
−log10 within−contrast BH FDR (cap50)
edgeR log2 fold change (case − control)
WT_21mo
vs WT_3mo
ETnERV2−int
5xFAD_3mo
vs WT_3mo
APOE4_TREM2_3mo
vs WT_3mo


Selected primary−contrast q<0.05 features; not independent validation
ETnERV2−int
WT_3mo_female_1
WT_3mo_female_2
WT_3mo_male_1
WT_3mo_male_2
WT_6mo_female_1
WT_6mo_female_2
WT_6mo_male_1
WT_6mo_male_2
WT_21mo_female_1
WT_21mo_female_2
WT_21mo_male_1
WT_21mo_male_2
5xFAD_3mo_female_1
5xFAD_3mo_female_2
5xFAD_3mo_male_1
5xFAD_3mo_male_2
APOE4_TREM2_3mo_female_1
APOE4_TREM2_3mo_female_2
APOE4_TREM2_3mo_male_1
APOE4_TREM2_3mo_male_2
Row z−score of primary TMM TE logCPM; display clipped at +/−2.5


WT_6mo
vs WT_3mo
−log10 within−contrast BH FDR (cap50)
edgeR log2 fold change (case − control)
WT_21mo
vs WT_3mo
−log10 within−contrast BH FDR (cap50)
edgeR log2 fold change (case − control)
5xFAD_3mo
vs WT_3mo
−log10 within−contrast BH FDR (cap50)
edgeR log2 fold change (case − control)
APOE4_TREM2_3mo
vs WT_3mo
−log10 within−contrast BH FDR (cap50)
edgeR log2 fold change (case − control)

Selected primary−contrast q<0.05 features; not independent validation

No features pass the declared selection